Two players take turns. Whoever takes the last counter wins, and a player who cannot move while counters are left loses. On a track, the token starts on the number of counters and moves toward 0 by the number of counters taken, and whoever puts it on 0 wins. When you play, two of you are the players and the third is the referee, and you switch roles after every game.

Problem 1: Take 1 or 2 counters on each turn. Play each pile with real counters until you are sure whether you would rather go first or second, and circle your choice. If you would rather go first, write how many counters you take on your first turn.

Pile of 7	first	second	take _____
Pile of 9	first	second	take _____
Pile of 11	first	second	take _____
Pile of 12	first	second	take _____
Pile of 16	first	second	take _____

Problem 2: For each rule, colour every square where you would like to leave the token for your opponent.

Take 1 or 2 counters.

0	1	2	3	4	5	6	7	8	9	10
	20	19	18	17	16	15	14	13	12	11

Take 1, 2 or 3 counters.

0	1	2	3	4	5	6	7	8	9	10
	20	19	18	17	16	15	14	13	12	11

Problem 3: Now take 1, 3 or 4 counters on each turn. Play each pile until you are sure whether you would rather go first or second, and circle your choice. If you would rather go first, write how many counters you take on your first turn.

Pile of 5	first	second	take _____
Pile of 8	first	second	take _____
Pile of 9	first	second	take _____
Pile of 11	first	second	take _____
Pile of 14	first	second	take _____

Problem 4: With moves of 1, 3 or 4, colour every square where you would like to leave the token for your opponent.

0	1	2	3	4	5	6	7	8	9	10
	20	19	18	17	16	15	14	13	12	11
	21	22	23	24	25	26	27	28	29	30

Problem 5: Kai plays with moves of 1, 3 or 4. He goes first and takes as many counters as he is allowed on his first turn. Find every pile from 1 to 30 where the player who goes first can win, but Kai's first move leaves a pile from which his opponent can win. Colour those piles.

0	1	2	3	4	5	6	7	8	9	10
	20	19	18	17	16	15	14	13	12	11
	21	22	23	24	25	26	27	28	29	30

Problem 6: A pile has 100 counters. Would you rather go first or second with moves of 1 or 2? With moves of 1, 2 or 3? With moves of 1, 3 or 4? Whenever you would go first, write how many counters you take. Explain how you would win whatever your opponent does.